IGCSE 0607 Practice Questions — E4.6

Perpendicular lines

Total Questions: 18 — **Total Marks:** 114 Marks

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  → **Paper 2 (Non-Calculator)** *(12 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*

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1 Line L has the equation $5y + 2x = 15$.
Find the gradient of a line that is perpendicular to L.

..... **[2]**

2 Find the equation of the line that is perpendicular to $y = 3x - 5$ and passes through the point $(6, 2)$. Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

3 Line L_1 has the equation $y = 5 - 2x$.
Line L_2 is perpendicular to L_1 and passes through the point $(4, 1)$.

(a) Find the gradient of line L_1.

..... **[1]**

(b) Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

4 Line L_1 has the equation $4y = 3x - 8$.

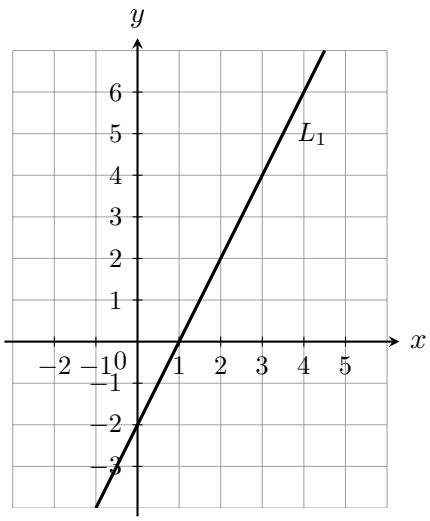
(a) Find the gradient of a line that is perpendicular to L_1.

Gradient = ... **[2]**

(b) Line L_2 is perpendicular to L_1 and passes through the point $(6, -5)$.
Find the equation of line L_2 in the form $y = mx + c$.

$y =$ ..... **[2]**

5 The diagram shows a straight line L_1 drawn on a coordinate grid.



(a) Write down the gradient of line L_1.

..... **[1]**

(b) Find the gradient of a line that is perpendicular to L_1.

..... **[1]**

(c) Line L_2 is perpendicular to L_1 and passes through the point $(4, 3)$.
Find the equation of line L_2 in the form $y = mx + c$.

$y =$ ..... **[3]**

6 A straight line L has the equation $2y = 5x + 8$.

(a) Find the gradient of line L.

..... **[2]**

(b) Find the gradient of a line that is perpendicular to line L.

..... **[1]**

(c) A line M is perpendicular to line L and passes through the point $(10, -3)$.
Find the equation of line M in the form $y = mx + c$.

$y =$ ..... **[3]**

7 Point A has coordinates $(-3, 8)$ and point B has coordinates $(9, -2)$.

(a) Find the coordinates of the midpoint of AB and the gradient of AB.

Midpoint = ... , Gradient = ... **[2]**

(b) Find the equation of the perpendicular bisector of AB.
Give your answer in the form $ax + by = c$, where a, b and c are integers.

..... **[3]**

8 A is the point $(-4, 7)$ and B is the point $(2, -1)$.

Find the equation of the perpendicular bisector of the line segment AB.
Give your answer in the form $ax + by = c$, where a, b and c are integers.

..... **[5]**

9 Line L has equation $2x + 3y = 12$.
Point A has coordinates $(-3, 8)$ and point B has coordinates $(9, -2)$.

(a) Find the gradient of a line that is perpendicular to line L.

..... **[2]**

(b) For the line segment joining $A(-3, 8)$ and $B(9, -2)$,

(i) find the coordinates of the midpoint of AB,

(..... , ...) **[2]**

(ii) find the equation of the perpendicular bisector of AB, giving your answer in the form $ax + by = c$, where a, b, and c are integers.

..... **[4]**

10 A is the point $(-3, 8)$ and B is the point $(9, -2)$.

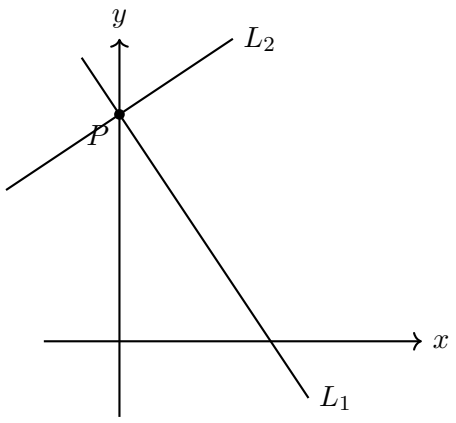
(a) Find the coordinates of the midpoint of AB.

(..... , ...) **[2]**

(b) Find the equation of the perpendicular bisector of AB.
Give your answer in the form $ax + by = c$, where a, b and c are integers.

..... **[4]**

11 Line L_1 has the equation $3x + 2y = 12$.
Line L_2 is perpendicular to L_1 and intersects L_1 at a point P on the y-axis.



Not to scale

(a) Find the coordinates of point P.

(..... , ...) **[1]**

(b) Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

(c) Point Q lies on L_2 and has an x-coordinate of 9.
Point R lies on L_1 and has an x-coordinate of 4.
Work out the area of triangle PQR.

..... **[5]**

12 Line L_1 has equation $kx + 2y = 12$.
Line L_2 passes through the points $A(0, 5)$ and $B(2, k + 2)$.
The constant k is a non-zero integer.

(a) Find the gradient of line L_1 in terms of k.

..... **[1]**

(b) Find the gradient of line L_2 in terms of k.

..... **[2]**

(c) Given that L_1 and L_2 are perpendicular, find the two possible values of k.

$k =$ ..... **[4]**

(d) Given that the gradient of line L_1 is positive, find the equation of line L_2 in the form $ax + by = c$, where a, b, and c are integers.

..... **[3]**

13 The equation of line P is $3x - 4y = 12$.
Line Q is perpendicular to line P and passes through the point $(6, -2)$.

Find the equation of line Q.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

14 The points A and B have coordinates $(-2, 4)$ and $(6, 0)$ respectively.

(a) Find the coordinates of the midpoint of AB.

(..... , ...) **[2]**

(b) Find the equation of the perpendicular bisector of the line segment AB.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

15 The coordinates of point A are $(-4, 5)$ and the coordinates of point B are $(2, -7)$.

(a) Find the coordinates of the midpoint of the line segment AB.

(..... , ...) **[2]**

(b) Calculate the gradient of the line AB.

..... **[2]**

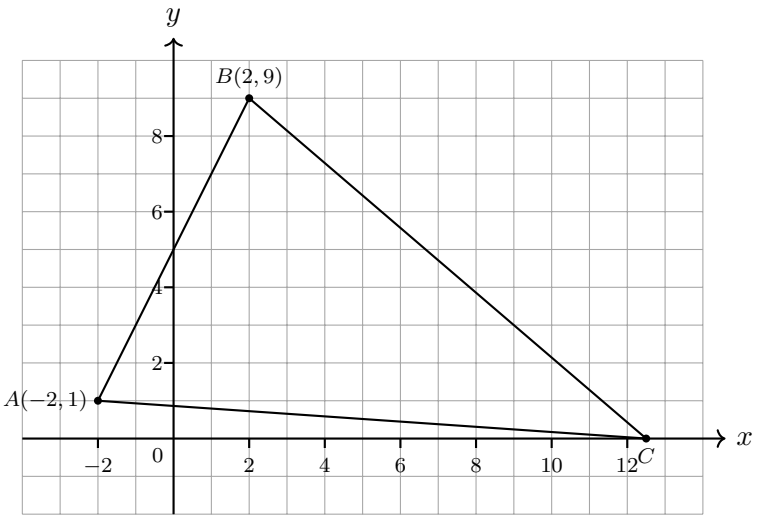
(c) Find the equation of the perpendicular bisector of the line segment AB.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

(d) The perpendicular bisector intersects the x-axis at point P.
Find the coordinates of P.

(..... , ...) **[2]**

16 The diagram shows a right-angled triangle ABC with $\angle ABC = 90°$.
The coordinates of A and B are $(-2, 1)$ and $(2, 9)$ respectively, and point C lies on the x-axis.



(a) Find the equation of the line BC.

$y =$ ..... **[3]**

(b) Find the coordinates of point C.

(..... , ...) **[2]**

(c) Calculate the area of triangle ABC.

Area = ... **[2]**

17 The line L_1 has equation $2y + 3x = 12$.
The line L_2 is perpendicular to L_1 and intersects L_1 at the point P on the y-axis.

(a) Find the coordinates of P.

(..... , ...) **[2]**

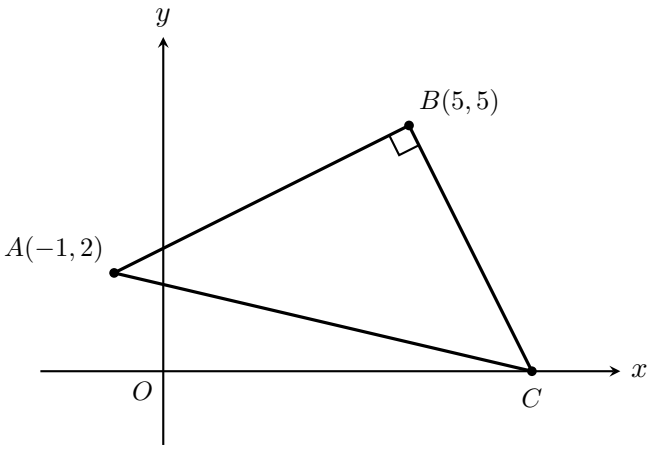
(b) Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

(c) Line L_2 crosses the x-axis at the point Q.
Calculate the exact length of PQ.

..... **[4]**

18 The diagram shows a sketch of triangle ABC with vertices $A(-1, 2)$ and $B(5, 5)$.
Angle $ABC = 90°$ and vertex C lies on the x-axis.



(a) Find the equation of the line passing through B and C in the form $y = mx + c$.

$y =$ ..... **[3]**

(b) Find the coordinates of vertex C.

$C =$ (..... , ...) **[2]**

(c) Calculate the area of triangle ABC.

..... square units **[3]**

(d) The perpendicular bisector of AC intersects the y-axis at point P.

Find the coordinates of P, giving the y-coordinate correct to 3 significant figures.

$P = (\ldots\ldots\ldots\ ,\ \ldots\ldots\ldots)$ **[4]**